

Module-1

Lecture-5 Deformation of Composite Sections

MECHANICS OF SOLIDS

- Analysis of bars of composite sections
- Problems
- Thermal stress
- Thermal stress in composite bars
- Problems

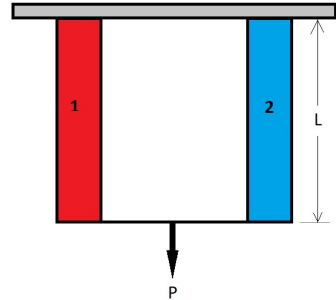
Analysis of bars of composite sections

Composite bar

A bar made of two or more bars of equal length but of different materials rigidly fixed with each other and behaving as one unit for expansion or compression when subjected to axial tensile or compressive force is called a **Composite bar**

For a composite bar two points are important

- Total external load on composite bar is equal to the sum of loads carried by each different materials
- Elongation of each bar is equal

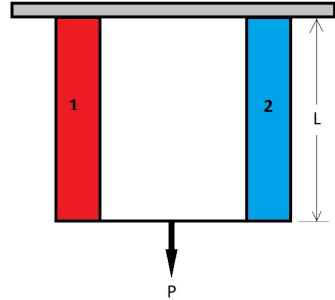


- $P = P_1 + P_2$
- $P_1 = \sigma_1 A_1$ $P_2 = \sigma_2 A_2$
- Elongation of each bar is equal. Since both bars have same length, Strain in each bars will also be same
- $\epsilon_1 = \frac{\sigma_1}{E_1}$ $\epsilon_2 = \frac{\sigma_2}{E_2}$
- $\epsilon_1 = \epsilon_2$

$$P = \sigma_1 A_1 + \sigma_2 A_2$$

$$\frac{\sigma_1}{E_1} = \frac{\sigma_2}{E_2}$$

The ratio of $\frac{E_1}{E_2}$ is called **Modular Ratio** of the first material to the second



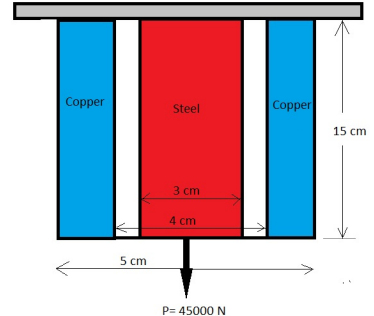
Workout 1

A steel rod of 3 cm dia is enclosed centrally in a hollow copper tube of external dia 5cm and internal dia 4 cm. The composite bar is then subjected to an axial pull of 45000 N. If the length of each bar is equal to 15cm, determine

- 1 The stress in rod and tube
- 2 Load carried by each bar

given: $E_{steel} = 0.21MN/mm^2$

$E_{copper} = 0.11MN/mm^2$

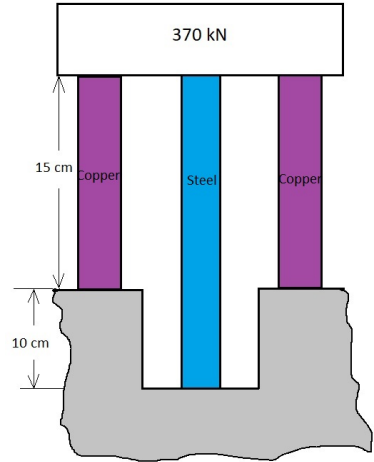


Workout 2

A steel rod and two copper rods together support a load of 370 kN as shown in figure. The cross sectional area of steel rod is 2500 mm^2 and of each copper rod is 1600 mm^2 . Find the stresses in the rods.

given: $E_{\text{steel}} = 2 \times 10^5 \text{ N/mm}^2$

$E_{\text{copper}} = 1 \times 10^5 \text{ N/mm}^2$



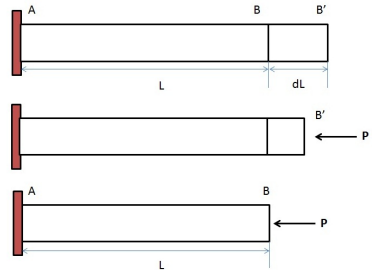
- ① A compound tube consist of a steel tube 140 mm internal and 160 mm external dia and an outer brass tube 160 mm internal and 180 mm external dia. The two tubes has same length of 140 mm. The compound tube carries an axial load of 900 kN. Find the stress and load carried by each tube and the amount it shortens. Take $E_{steel} = 0.2MN/mm^2$
 $E_{brass} = \times 10^5 N/mm^2$
- ② Three bars made of copper, zinc and aluminium are of equal length and have cross section 500, 750, and 1000 mm^2 respectively. They are rigidly connected at their ends. If the compound member is subjected to a longitudinal pull of 250 kN, estimate the proportional of the load carried on each rod and the induces stress. Take $E_{copper} = 1.3 \times 10^5 N/mm^2$
 $E_{zinc} = 0.1MN/mm^2$ $E_{aluminium} = 0.8 \times 10^5 N/mm^2$

Thermal Stresses

Thermal Stresses

- Thermal stresses are the stresses induced in a body due to change in temperature
- Thermal stresses are set up in a body, when temperature is raised or lowered and the body is not allowed to expand or contract freely
- Free extension of the rod, $dL = \alpha TL$
- Compressive strain $= \alpha T$

$$\text{Thermal Strain, } e = \frac{\text{Extension Prevented}}{\text{Original Length}} = \alpha T$$
$$\text{Thermal Stress, } \sigma = \alpha TE$$



*Thank
you!*